

Day 3 — Fifteen Tithis per Pakṣa

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recite the 15 tithi names (pratipadā, dvitīyā, ...) and explain why a tithi isn't exactly a day.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *śukla pakṣa* (IAST: śukla pakṣa; lit. “waxing fortnight (new → full)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Define tithi precisely.** A *tithi* is the time taken for the Moon to gain 12° of longitudinal separation from the Sun. 30 tithis per synodic month → 15 per pakṣa. A tithi is therefore a unit of angular progress, NOT a unit of time.
2. **Why a tithi ≠ a day.** Lunar speed varies (elliptical orbit, ~3.4°/hour to ~5.5°/hour). A tithi can last 19 to 26 hours. Two consecutive tithis can both touch the same sunrise (*kṣaya tithi*) or one tithi can span two sunrises (*adhika tithi*).
3. **Name the 15 tithis.** *pratipadā, dvitīyā, tṛtīyā, caturthī, pañcamī, ṣaṣṭhī, saptamī, aṣṭamī, navamī, daśamī, ekādaśī, dvādaśī, trayodaśī, caturdaśī, pūrṇimā/amāvasyā*. The 15th is *pūrṇimā* in śukla pakṣa, *amāvasyā* in kṛṣṇa pakṣa.
4. **Source.** Sūrya-siddhānta, ch. 2 (lunar motion); also *Vasiṣṭha-siddhānta* tradition.
5. **Computation exercise.** Given today’s Moon-Sun elongation in degrees (provide one), students compute today’s tithi number. Then compare with the published pañcāṅga.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Fifteen Tithis per Pakṣa.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).